

Sydney Girls High School



Mathematics Extension 1

Year 11

Assessment Task 3

2017

Reading time : 5 minutes

Writing time : 60 minutes

Total marks : 60

Topics: Other inequalities, Internal/External Division of an Interval, 3 Dimensional Trigonometry, the Quadratic Polynomial, Locus, Further Trigonometry (excluding Integration), Geometrical application of Differentiation, Series and Series Applications

Instructions:

- There are FOUR (4) questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write using a black or blue pen.

Student 's Name: _____ **Teacher's Name:** _____

Question 1**15 Marks**

(a) What is the 20th term of the sequence defined by $T_n = (n-1)(3+n)$? **1**

(b) Given the equation of the parabola $(x-3)^2 = 16(y+2)$, find:

(i) the coordinates of the vertex . **1**

(ii) the coordinates of the focus. **1**

(iii) the equation of the axis of symmetry. **1**

(iv) the equation of the directrix. **1**

(c) The quadratic equation $3x^2 - 2x - 5 = 0$ has roots α and β . Evaluate :

(i) $\alpha + \beta$ **1**

(ii) $\alpha\beta^2 + \beta\alpha^2$ **1**

(iii) $\alpha^2 + \beta^2$ **2**

(d) Find the coordinates of the point that divides the interval joining the points **3**

(0,4) and (2,7) externally in the ratio 2 : 3.

(e) Solve $\frac{2}{x+1} > 4$. **3**

- (a) Find the obtuse angle between the lines $y = 2x - 1$ and $3x - 2y + 1 = 0$.
Give your answer correct to the nearest degree.

3

- (b) Find the limiting sum of the series $8 - 4 + 2 - 1 + \dots$

2

- (c) Find A , B and C such that

$$3x^2 - x + 3 = A(x - 1)(x + 2) + B(x - 2) + C.$$

3

- (d) Find the primitive function of $4x - \frac{1}{x^2}$.

2

- (e) Given $2\sin^2 \theta = 3\cos \theta$, find all possible values of θ where $0^\circ \leq \theta \leq 360^\circ$.

3

- (f) Given that $\tan A = 2 \tan B + \cot B$, prove the following identity:

$$2 \tan(A - B) = \cot B$$

2

- (a) A and B are the points $(4, 2)$ and $(1, 5)$ respectively. Find the equation of the locus of a point $Q(x, y)$ which moves such that QA is perpendicular to QB . 2
- (b)
- (i) Express $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. 2
- (ii) Hence, or otherwise, solve the equation $\sin x + \sqrt{3} \cos x = \sqrt{2}$ for $0^\circ \leq x \leq 360^\circ$. 2
- (c) For what values of k does the quadratic equation $kx^2 - 4kx - k + 5 = 0$ have real, distinct roots? 2
- (d) Consider the curve $y = x^3 - x^2 - 5x + 1$
- (i) Find the coordinates of any turning points and find their nature. 3
- (ii) Find any points of inflexion. 1
- (iii) Sketch the curve. You do not need to find the x -intercepts. 2
- (iv) For what values of x is the curve both decreasing and concave up? 1

Question 4 (Start a new page)**15 marks**

- (a) An open cylindrical can, without the top lid, has an external surface area of $20\pi \text{ cm}^2$.

- (i) Let r be the radius and h be its height in centimetres. Show that

1

$$h = \frac{20 - r^2}{2r}.$$

- (ii) Hence, show that the volume of the can is given by

1

$$V = 10\pi r - \frac{1}{2}\pi r^3.$$

- (iii) Show that the maximum volume is obtained when the height of the can is equal to its radius.

3

- (b) A surveyor observes two towers A and B . Tower A is due north of the surveyor with a height of 80 m, and Tower B is on a bearing of θ (where $\theta < 90^\circ$) from the surveyor with a height of 120 m. The angles of elevation of the tops of the towers A and B are 40° and 36° respectively from the surveyor. The towers are 150 m apart on a horizontal plane. Calculate the value of θ to the nearest minute.

3

- (c) Mathew borrowed \$100000 to buy an apartment, which he is going to rent. He plans to repay \$500 every fortnight from his salary. On every 4th week he will repay an extra \$900 from the rent he receives. Interest on the loan is 6.5% p.a. reducing fortnightly.

Let A_n is the amount owing on the loan after n fortnights and $R=1.0025$

- (i) Show that $A_2 = 100000R^2 - 500(R + 1) - 900$

1

- (ii) Show that

2

$$A_4 = 100000R^4 - 500(R^3 + R^2 + R + 1) - 900(R^2 + 1)$$

- (iii) Hence show that $A_{100} = 100000R^{100} - 500\frac{R^{100}-1}{R-1} - 900 \times \frac{R^{100}-1}{R^2-1}$

2

- (iv) What percentage of the money paid in first 100 repayments will be interest?

2**END OF PAPER**

Q1

$$\begin{aligned} \text{a) } T_{20} &= (20-1)(3+20) \\ &= 437 \checkmark \end{aligned}$$

$$\text{b) } (x-3)^2 = 16(y+2)$$

$$\text{i) } V(3, -2) \checkmark$$

$$\begin{aligned} \text{ii) } 4a &= 16 \\ a &= 4 \end{aligned}$$

Focus point $S(x_0, y_0 + a)$

$$S(3, 2) \checkmark$$

$$\text{iii) } x = 3 \checkmark$$

$$\text{iv) } \text{Directrix: } y = -6 \checkmark$$

$$\text{c) } 3x^2 - 2x - 5 = 0$$

$$\text{i) } \alpha + \beta = \frac{2}{3} \checkmark$$

$$\text{ii) } \alpha\beta = -5/3 \checkmark$$

$$\alpha\beta^2 + \beta\alpha^2 = \alpha\beta(\alpha + \beta)$$

$$= -\frac{5}{3} \left(\frac{2}{3} \right)$$

$$= -\frac{10}{9} \checkmark$$

$$\text{iii) } (\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \checkmark$$

$$= \frac{4}{9} + \frac{10}{3} = \frac{34}{9} \checkmark$$

Q1

d) $A(0, 4)$ $B(2, 7)$. External Division
ratio $2:3$

Let $P(x, y)$ be the point that divides
the interval.

$$x = \frac{0(-3) + 2(2)}{2 + -3} = -4$$

$$y = \frac{4(-3) + 7(2)}{2 + -3} = -2$$

$$P(-4, -2)$$

* A number of students
could not recognise
the difference between
Internal and External
Ratio Division.

e) $\frac{2}{x+1} > 4$

$$2(x+1) > 4(x+1)^2$$

$$4(x+1)^2 - 2(x+1) < 0$$

$$2(x+1) [2(x+1) - 1] < 0$$

$$2(x+1)(2x+1) < 0$$

$$-1 < x < -\frac{1}{2}$$

$$2(a) m_1 = 2, m_2 = \frac{3}{2}$$

$$\tan \theta = \frac{2 - \frac{3}{2}}{1 + 2 \times \frac{3}{2}}$$

$$\theta = 7.1250163 + 9$$

$$= 7^\circ$$

$$\text{obtuse } \angle = 180^\circ - 7^\circ$$

$$= 173^\circ$$

Many students used the wrong formula.

Many students forgot what an obtuse $\angle$ is.

$$(h) S_\infty = \frac{8}{1 - -\frac{1}{2}}$$

$$= \frac{16}{3}$$

$$(c) 3x^2 - x + 3 = A(x^2 + 2x - 2) + Bx - 2B + C$$

$$= Ax^2 + (A + 2B)x - 2B + C - 2A$$

$$A = 3, A + 2B = -1 \quad -2B + C - 2A = 3$$

$$3 + 2B = -1 \quad 8 + C - 6 = 3$$

$$B = -4 \quad C = 1$$

$$(d) 4x - x^{-2}$$

Primitive function

$$\frac{4x^2}{2} - \frac{x^{-1}}{-1} + c = 2x^2 + \frac{1}{x} + c$$

$$(e) 2(1 - \cos^2 \theta) - 3 \cos \theta = 0$$

$$2 - 2 \cos^2 \theta - 3 \cos \theta = 0$$

$$2 \cos^2 \theta + 3 \cos \theta - 2 = 0$$

$$2 \cos^2 \theta + 4 \cos \theta - \cos \theta - 2 = 0$$

$$2 \cos \theta (\cos \theta + 2) - 1(\cos \theta + 2) = 0$$

$$(2 \cos \theta - 1)(\cos \theta + 2) = 0$$

$$\cos \theta = \frac{1}{2} \quad \cos \theta = -2$$

$$\therefore \theta = 60^\circ, 300^\circ$$

$$(f) 2 \tan(A - B)$$

$$= 2 \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$= 2 \frac{2 \tan B + \cot B - \tan B}{1 + \tan A \tan B}$$

$$= 2 \frac{\tan B + \cot B}{1 + (2 \tan B + \cot B) \tan B}$$

$$= 2 \frac{\tan B + \cot B}{1 + 2 \tan^2 B + 1}$$

$$= 2 \frac{(\tan^2 B + 1) \cot B}{2 + 2 \tan^2 B}$$

$$= \frac{(1 + \tan^2 B) \cot B}{1 + \tan^2 B}$$

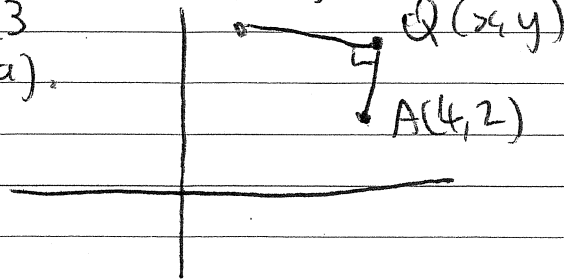
$$= \cot B$$

Students who used the A method were generally unable to complete the process correctly.

2017 YEAR 11 EXT TASK 3.

Q3

(a).



If $QA \perp QB$ then $m_{QA} \times m_{QB} = -1$

$$m_{QA} = \frac{y-2}{x-4}$$

$$m_{QB} = \frac{y-5}{x-1}$$

$$\frac{y-2}{x-4} \times \frac{y-5}{x-1} = -1$$

$$(y-2)(y-5) = -(x-4)(x-1)$$

$$y^2 - 7y + 10 = -x^2 + 5x - 4$$

$$x^2 - 5x + y^2 - 7y + 14 = 0$$

(b) (i) $R \sin(x+\alpha) = R \cos \alpha \sin x + R \sin \alpha \cos x$

$$R \cos \alpha = 1 \quad R \sin \alpha = \sqrt{3}$$

$$R^2 = 1+3$$

$$\tan \alpha = \sqrt{3}$$

$$R = 2$$

$$\alpha = 60^\circ$$

$$\sin x + \sqrt{3} \cos x \equiv 2 \sin(x+60^\circ)$$

(b)(ii). $\sin x + \sqrt{3} \cos x = \sqrt{2}$

$$2 \sin(x + 60^\circ) = \sqrt{2}$$

$$\sin(x + 60^\circ) = \frac{1}{\sqrt{2}}$$

When you lose a solution at the lower end of the domain you gain one at the upper end.

$$x + 60^\circ = 45^\circ, 135^\circ, 405^\circ$$

$$x = -15^\circ, 75^\circ, 345^\circ$$

So $x = 75^\circ, 345^\circ$ since $0^\circ \leq x \leq 360^\circ$

(c) $\Delta > 0$

For distinct roots the discriminant can not equal zero.

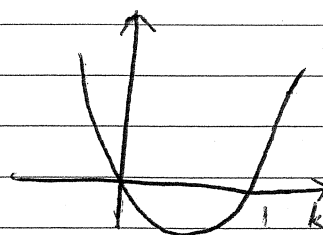
$$16k^2 - 4k(-k + 5) > 0$$

$$16k^2 + 4k^2 - 20k > 0.$$

$$20k^2 - 20k > 0$$

$$k^2 - k > 0.$$

$$k(k - 1) > 0$$



$$k < 0, k > 1.$$

(d)(i) $y = x^3 - x^2 - 5x + 1$

$$\frac{dy}{dx} = 3x^2 - 2x - 5.$$

$$\frac{d^2y}{dx^2} = 6x - 2.$$

Stat Pts $\frac{dy}{dx} = 0$

$$3x^2 - 2x - 5 = 0$$

$$\frac{(3x-5)(3x+3)}{3} = 0$$

$$(3x-5)(x+1) = 0$$

$$x = \frac{5}{3}, -1.$$

Nature
at $x = \frac{5}{3}$

$$\frac{d^2y}{dx^2} = 6\left(\frac{5}{3}\right) - 2 = 8 > 0 \text{ minima}$$

at $x = -1$

$$\frac{d^2y}{dx^2} = 6(-1) - 2 = -8 < 0 \text{ maxima.}$$

Points
at $x = \frac{5}{3}$ $y = -\frac{148}{27} \approx -5.5.$

at $x = -1$ $y = 4.$

Turning Points $\left(\frac{5}{3}, -\frac{148}{27}\right)$ minima.

$(-1, 4)$ maxima.

(ii) Inflection point

$$\frac{d^2y}{dx^2} = 0.$$

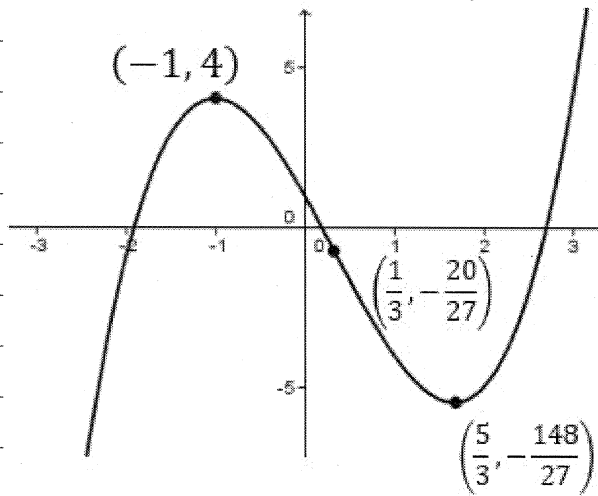
$$6x - 2 = 0.$$

$$x = \frac{1}{3}.$$

$$y = -\frac{20}{27} \approx -0.7$$

Inflection $(\frac{1}{3}, -\frac{20}{27})$.

(iii)



(iv) $\frac{1}{3} < x < \frac{5}{3}$.

Many students included the end points with less than or equal to.

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4) a)

$$\begin{aligned} \text{i) } SA &= \pi r^2 + 2\pi rh \\ 20\pi &= \pi r^2 + 2\pi rh \\ 20 &= r^2 + 2rh \\ 2rh &= 20 - r^2 \\ h &= \frac{20 - r^2}{2r} \end{aligned}$$

$$\begin{aligned} \text{ii) } V &= \pi r^2 h \\ &= \pi (r^2) \left(\frac{20 - r^2}{2r} \right) \\ &= \pi \left(\frac{20r - r^3}{2} \right) \\ &= \pi \left(10r - \frac{r^3}{2} \right) \\ &= 10\pi r - \frac{\pi r^3}{2} \end{aligned}$$

$$\text{iii) } V' = 10\pi - \frac{3\pi r^2}{2}$$

$$\begin{aligned} V' &= 0 \\ 20\pi &= 3\pi r^2 \end{aligned}$$

$$r^2 = \frac{20}{3}$$

$$r = \sqrt{\frac{20}{3}}$$

$$V'' = -3\pi r$$

< 0

$$\therefore \text{max}$$

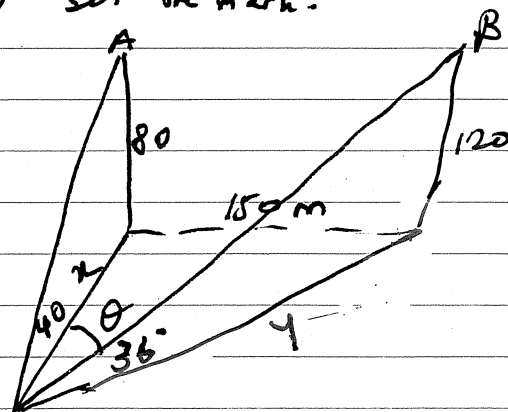
$$r = \sqrt{\frac{20}{3}}$$

$$h = \frac{20 - \frac{20}{3}}{2\left(\sqrt{\frac{20}{3}}\right)}$$

* If V'' not shown no marks awarded

$$\begin{aligned} &= \frac{\frac{40}{3}}{\frac{2\sqrt{20}}{\sqrt{3}}} \\ &= \frac{20\sqrt{3}}{3\sqrt{20}} \times \frac{\sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{20}}{\sqrt{20}} \\ &= \frac{20 \times 3 \sqrt{20}}{3 \sqrt{3} \times 20} \\ &= \frac{\sqrt{20}}{\sqrt{3}} = r \end{aligned}$$

Many students couldn't show this step if you didn't rationalise you couldn't set the mark.



$$\tan 40 = \frac{80}{x}$$

$$x = \frac{80}{\tan 40}$$

$$\tan 36 = \frac{120}{y}$$

$$y = \frac{120}{\tan 36}$$

$$\cos \theta = \frac{\left(\frac{120}{\tan 36}\right)^2 + \left(\frac{80}{\tan 40}\right)^2}{2 \times \left(\frac{120}{\tan 36}\right) \left(\frac{80}{\tan 40}\right)} = 150^2$$

$$\theta = 63^\circ 52'$$

c)

$$i) A_1 = 100000(R) - 500$$

$$A_2 = (100000R - 500)R - 500 - 900$$

$$= 100000R^2 - 500R - 500 - 900$$

$$= 100000R^2 - 500(R+1) - 900$$

* Many students couldn't show these steps properly. so no marks were awarded if steps not shown.

$$ii) A_3 = (100000R^2 - 500(R+1) - 900)R - 500$$

$$= 100000R^3 - 500(R^2 + R) - 900R - 500$$

$$= 100000R^3 - 500(R^2 + R + 1) - 900R$$

$$A_4 = (100000R^3 - 500(R^2 + R + 1) - 900R)R - 500 - 900$$

$$= 100000R^4 - 500(R^3 + R^2 + R) - 900R^2 - 500 - 900$$

$$= 100000R^4 - 500(R^3 + R^2 + R + 1) - 900(R^2 + 1)$$

↓ xib you didn't show $(R^2)^{50}$ no mark.

$$iii) A_{100} = 100000R^{100} - 500(R^{99} + R^{98} + \dots + 1) - 900(R^{98} + R^{96} + \dots + 1)$$

$$= 100000R^{100} - 500 \left(\frac{1(R^{100} - 1)}{R - 1} \right) - 900 \left(\frac{1(R^{100} - 1)}{R^2 - 1} \right)$$

$$= 100000R^{100} - 500 \left(\frac{R^{100} - 1}{R - 1} \right) - 900 \left(\frac{R^{100} - 1}{R^2 - 1} \right)$$

$$iv) A_{100} = \$20648.77 \quad \text{paid } \$79351.23$$

$$\text{Actually paid} = 500 \times 100 + 50 \times 900 = 95000$$

$$I\% = \frac{15648.77}{95000} \times 100\% = 16.47\%$$

* Many students had problem with this equation.